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Q-3

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Q.1 Outer loop: $i=1, 2, 3, \dots, N$.

Inner loop condition is:

$$ij \leq N$$

$$j \leq \frac{N}{i} \text{ times}$$

So, Total is:

$$\sum_{i=1}^N \frac{N}{i} = N \sum_{i=1}^N \frac{1}{i}$$

Means, for $\sum_{i=1}^N \frac{1}{i}$ means $\frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^m}$

$$\text{or } \sum_{i=1}^N \frac{1}{i} = O(\log N).$$

$$\text{Therefore } = O(N \log N).$$

$$\boxed{T.C = O(N \log N)}$$

Q2 Outer loop runs N times.for every i inner loop condition is:

$$j^2 \leq i$$

$$\text{Therefore } j \leq \sqrt{i}$$

So inner loop has Time Complexity $O(\sqrt{i})$.

So, for Total loop:

$$\sum_{i=1}^N \sqrt{i}$$

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Using the standard summation

$$\sum_{i=1}^N \sqrt{i} = O(N^{3/2})$$

While second loop takes.

$$O(N).$$

Since; $N^{3/2} > N.$

So

$$T.C = O(N^{3/2}).$$

Q3: Outer loop: $i=1$

$i=2.$

$i=4.$

$i=8$ and so on.

means it double every time.

i.e. $2^k = n.$

$$k \log_2 = \log n.$$

$$O(\log n).$$

Inner loop: $j=1.$

$j=3.$

$j=9.$

$j=27. \dots$

So it triplet every time.

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Number of iterations is;

$$3^k = n$$

$$k \log_3 = \log n$$

$$O(\log_3 n)$$

Therefore $O(\log n)$.

Nested loops So; $O(\log N) \times O(\log N)$

Hence;

$$T.C = O[(\log n)^2] \text{ or } T.C = O(\log^2 n)$$

Q4 The loop is;

for (long long i = 2; i < N; i = i * i).

So; values are:

$$2 \rightarrow 2^2 \rightarrow 4^2 = 16 \rightarrow 16^2 \rightarrow 256 \rightarrow 256^2 \dots$$

So;

$$i = 2^{2^k} \text{ after } k \text{ iterations.}$$

loop stops when:

$$2^{2^k} > N$$

$$\text{So; } \log \log N = 2^k$$

$$\log \log N = k$$

So;

$$T.C = O(\log \log N)$$